

Introduction

During the internship, three comprehensive Power BI dashboards were developed to gain practical experience in Business Intelligence, data modeling, data visualization, and analytical reporting. The tasks involved importing datasets, performing data preprocessing, creating data models, writing DAX measures, designing interactive dashboards, and validating business insights. These activities strengthened practical knowledge of Microsoft Power BI and enhanced skills in transforming raw data into meaningful visual reports.

1. Sales Performance Dashboard

Objective

The objective of this task was to build an interactive Sales Performance Dashboard that provides real-time insights into sales, profit, customer orders, and regional performance. The dashboard enables business users to monitor key performance indicators and make informed decisions through interactive visualizations.

Work Performed

The following activities were completed during the development of the Sales Performance Dashboard:

- Imported the Global Superstore dataset into Power BI Desktop.
- Performed data cleaning using Power Query.
- Removed unnecessary columns and corrected data types.
- Replaced missing values to improve data quality.
- Created a Star Schema consisting of Fact and Dimension tables.
- Established relationships among Product, Customer, Geography, Calendar, and Sales tables.
- Developed DAX measures for business KPIs.
- Created KPI cards displaying important business metrics.
- Designed interactive visualizations including maps, bar charts, treemaps, line charts, and tables.
- Added slicers for dynamic filtering.
- Implemented drill-down and drill-through functionality.
- Applied dashboard themes, formatting, and alignment.
- Tested dashboard functionality and validated the generated reports.

Skills Applied

- Data Import
- Data Cleaning
- Power Query
- Data Modeling
- Star Schema Design
- DAX Measures
- Interactive Dashboard Design
- Business Intelligence Reporting

Outcome

A professional Sales Performance Dashboard was successfully developed, allowing users to analyze sales trends, product performance, regional sales, and profitability through interactive visual reports.

2. Customer Insights Dashboard

Objective

The objective of this task was to analyze customer behavior, customer segmentation, customer lifetime value, and customer retention using interactive Power BI dashboards. The dashboard supports marketing and sales teams by providing meaningful customer insights.

Work Performed

The following activities were completed:

- Imported and cleaned customer-related data.
- Built a customer-centric Star Schema.
- Created DAX measures for customer analytics.
- Calculated:
 - Total Customers
 - Customer Lifetime Value
 - Average Revenue per Customer
 - New Customers
 - Returning Customers
- Created customer segmentation using DAX.
- Developed KPI Cards.

- Built Customer Segment visualizations.
- Created Top Customers analysis.
- Added slicers for customer filtering.
- Configured customer drill-through pages.
- Applied dashboard formatting and testing.

Skills Applied

- Customer Analytics
- Data Modeling
- DAX Calculations
- Dashboard Development
- Data Visualization
- Interactive Reporting

Outcome

The Customer Insights Dashboard successfully provided customer segmentation, purchasing behavior, revenue contribution, and customer retention analysis, enabling better customer-focused decision making.

3. Inventory & Shipping Analytics Dashboard

Objective

The objective of this task was to analyze logistics performance, shipping efficiency, delivery time, product profitability, and shipping costs using Power BI. The dashboard supports operational decision-making through interactive visualizations.

Work Performed

The following activities were completed:

- Imported logistics and shipping datasets.
- Cleaned and transformed the data.
- Created Star Schema relationships.
- Developed DAX measures including:
 - Total Shipping Cost
 - Average Shipping Cost
 - Average Delivery Days
 - Shipping Cost Percentage
 - Profit per Product

- Designed KPI Cards.
- Created Shipping Cost Analysis.
- Developed Product Profitability Scatter Chart.
- Built Shipping Trend Analysis.
- Created Top and Bottom Product reports.
- Added slicers and drill-through pages.
- Performed dashboard testing and validation.

Skills Applied

- Logistics Analytics
- Shipping Performance Analysis
- Data Modeling
- DAX Measures
- Business Dashboard Design
- Data Visualization

Outcome

The Inventory and Shipping Analytics Dashboard successfully analyzed shipping costs, delivery performance, and product profitability, helping identify opportunities for operational improvements.

Overall Learning Outcomes

Throughout these Power BI tasks, the following practical skills were developed:

- Data Import and Transformation
- Data Cleaning using Power Query
- Data Modeling using Star Schema
- Relationship Creation
- DAX Measure Development
- KPI Dashboard Design
- Interactive Data Visualization
- Slicer and Filter Implementation
- Drill-Down and Drill-Through Analysis
- Business Intelligence Reporting
- Dashboard Testing and Validation
- Data-Driven Decision Support

Conclusion

The successful completion of these Power BI dashboard development tasks provided practical exposure to Business Intelligence concepts and modern data analytics techniques. By integrating data preprocessing, modeling, DAX calculations, and visualization, professional dashboards were created to analyze sales performance, customer behavior, and inventory logistics. These tasks significantly enhanced practical knowledge of Power BI and strengthened analytical, problem-solving, and reporting skills applicable to real-world business environments.